

Notes on: Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning

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Problem

Applying supervised ML to science is unreasonable due to the high cost of training data.

Key idea

Substitute the supervised approach with unsupervised or semi-supervised one. This is possible due to our knowledge of the governing equations.

Construct a parametrized function and plug it into the differential equation. Minimize the residual via gradient descent.

What I'd prove/extend